

Pythagoras' Theorem & Trigonometry

Extra practice test 3 — answer sheet · 27 marks total

Section A — Pythagoras' theorem

A1. $a^2 + a^2 = 18^2 \rightarrow 2a^2 = 324$ (1) $a^2 = 162 \rightarrow a = \sqrt{162} = \mathbf{12.7 \text{ cm}}$ (1)

A2. Splitting the triangle in half gives a right-angled triangle with hypotenuse 16 cm and base 8 cm (1)
 $\text{height}^2 = 16^2 - 8^2 = 256 - 64 = 192$ (1) $\text{height} = \sqrt{192} = \mathbf{13.9 \text{ cm}}$ (1)

A3. $11^2 + 60^2 = 121 + 3600 = 3721$ (1) $61^2 = 3721$ — since these are equal, the triangle is **right-angled** (1)

Section B — SOH CAH TOA basics

B1. **True** — opposite = hypotenuse $\times \sin \theta$, and $\sin \theta$ increases as θ increases from 0° to 90° , so a larger θ gives a larger opposite side (1)

B2. **Greater than 1** — $\tan 45^\circ = 1$, and $\tan \theta$ keeps increasing as θ increases towards 90° , so $\tan 80^\circ > 1$ (1)

B3. $\sin 70^\circ = 0.94$, $\cos 70^\circ = 0.34$ — **$\sin 70^\circ$ is larger** (1)

Section C — Finding missing sides

C1. $\cos 40^\circ = x \div 22$ (1) $x = 22 \times \cos 40^\circ$ (1) = **16.9 cm** (1)

C2. $\tan 51^\circ = h \div 9 \rightarrow h = 9 \times \tan 51^\circ$ (1) = **11.1 m** (1)

C3. $\tan 24^\circ = 5 \div x \rightarrow x = 5 \div \tan 24^\circ$ (1) = **11.2 cm** (1)

Section D — Finding missing angles

D1. $\tan \theta = 18 \div 150 = 0.12$ (1) $\theta = \tan^{-1}(0.12) = \mathbf{6.8^\circ}$ (1)

D2. $\cos \theta = 11 \div 19 = 0.5789$ (1) $\theta = \cos^{-1}(0.5789)$ (1) = **54.6°** (1)

Section E — Applying Pythagoras & trigonometry

E1. $\cos \theta = 1.5 \div 4 = 0.375 \rightarrow \theta = \cos^{-1}(0.375) = 68.0^\circ$ (1) since $68.0^\circ > 65^\circ$, the ladder **meets the safety requirement** (1)

E2. (a) $PR^2 = 15^2 + 9^2 = 225 + 81 = 306 \rightarrow PR = \sqrt{306} = \mathbf{17.5 \text{ km}}$ (1) (b) $\tan \theta = 9 \div 15 = 0.6$ (1) $\theta = \tan^{-1}(0.6) = \mathbf{31.0^\circ}$ (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any correct rounding to 1 decimal place unless a different level of accuracy is specified. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.